

components q_i, p_i, Q_i, P_i , respectively; we may think of Q_i and P_i as functions of $\mathbf{q}$ and $\mathbf{p}$, then

$$dQ_i = dq_i + hm_i^{-1}dP_i, \quad (2.20)$$

$$dP_i = dp_i - h \sum_{j=1}^{N_c} \frac{\partial^2 U}{\partial q_j \partial q_i} dq_j. \quad (2.21)$$

Computing the 2-form,

$$dQ_i \wedge dP_i = dq_i \wedge dP_i + hm_i^{-1}dP_i \wedge dP_i,$$

but $du \wedge du \equiv 0$ for any u , hence

$$dQ_i \wedge dP_i = dq_i \wedge dP_i,$$

and (2.21) implies

$$dQ_i \wedge dP_i = dq_i \wedge dp_i - h \sum_{j=1}^{N_c} dq_i \wedge \frac{\partial^2 U}{\partial q_j \partial q_i} dq_j.$$

It is then a simple exercise to show that

$$\sum_{i=1}^{N_c} \sum_{j=1}^{N_c} dq_i \wedge \frac{\partial^2 U}{\partial q_j \partial q_i} dq_j = 0,$$

using the skew-symmetry of the wedge product and the fact that the Hessian matrix is symmetric. This implies that

$$\sum_{i=1}^{N_c} dQ_i \wedge dP_i = \sum_{i=1}^{N_c} dq_i \wedge dp_i,$$

which means that the method is symplectic.

2.3.7 The Adjoint Method

Given any numerical integrator $\mathcal{G}_h$, consider the map

$$\mathcal{G}_h^\dagger = \mathcal{G}_{-h}^{-1}.$$